

Abhinav Jagan Polimera

abhinavpolimera@gmail.com | LinkedIn | GitHub | Portfolio

U.S. Citizen | No sponsorship required | Open to relocation

Profile: Software engineer across AI applications, developer infrastructure, distributed systems, and networking. Drawn to high-ownership problems where reliability, scale, and low-level constraints shape the product.

Education

Birla Institute of Technology and Science, Pilani - Hyderabad Campus

2020-2024

B.E. in Computer Science | CGPA: 8.2/10

Work Experience

Cisco Systems

On-site

Software Developer 2 | Distributed Systems

Apr 2026 - Present

- Built a semantic knowledge graph for Cisco Silicon One router OS code, improving LLM retrieval precision by about 30% while reducing token use.
- Co-developed an AI infrastructure prototype running 0.5-2 MB quantized models on Cisco routers at under 2% CPU, using telemetry for model selection and advancing it to customer evaluation.
- Delivered ERSpan over GUE for Microsoft, cutting tunnel overhead 16%, avoiding about 5 PB/year of telemetry traffic, and increasing monitoring-fabric capacity by 2.3%.

Software Developer | Distributed Systems

Aug 2024 - Apr 2026

- Led Cisco 8000 egress traffic mirroring for SoftBank's migration, matched NCS 5000 behavior, and cut delivery time by 60%.
- Built AI-assisted developer tools used by 100+ engineers, including metrics analytics and a traffic-mirroring simulator with MCP modules; cut issue reproduction from five days to one.
- Owned SPAN, Lawful Intercept, and NetFlow delivery and debugging across Cisco router platforms.

Technical Intern | Distributed Systems

Jan 2024 - Jun 2024

- Automated NTP, VRF, and OpenConfig validation in Python (88% coverage) and parallelized router-OS tests, cutting runtime 90% and earning a Cisco excellence award.

Projects and Research

X-Agent - Local Multi-Agent Content Workflow

GitHub

- Built a local-first LangGraph/FastAPI agent workflow with Ollama, local embeddings, retrieval, critique, approval gates, Docker, automated tests, and SSRF/DoS safeguards.

UAV-Based Suspicious-Activity Detection

IEEE

- Co-authored an IEEE paper on a Python/YOLO UAV surveillance system using object counts and activity duration to flag suspicious actions at remote locations.

NeRF and Photogrammetry for 3D Reconstruction

ICAART

- First-authored an ICAART 2024 comparison of NeRF and photogrammetry against ground-truth object dimensions; NeRF reached up to 99.89% accuracy on the evaluated geometric attributes.

Technical Skills

Languages: Python, C, C++, Shell

AI/ML: Agentic workflows, RAG, knowledge graphs, embeddings, LangGraph, MCP, Ollama, model quantization

Systems & CV: Distributed systems, Linux, network OS, routing, Cisco Silicon One, telemetry, packet processing, FastAPI, Docker, OpenCV, YOLO, NeRF

Interests and Activities

Creative: Self-taught photographer; sketching, digital art, 3D printing, automotive design, F1, and sim racing.

Activities: School chess representative; karate blue belt; 13,000+ ft trek; Vietnam cave hiking and backpacking.